

ID N1**Function and characteristics of a PSD-based distance sensor (SEN-IR)****Required Knowledge**

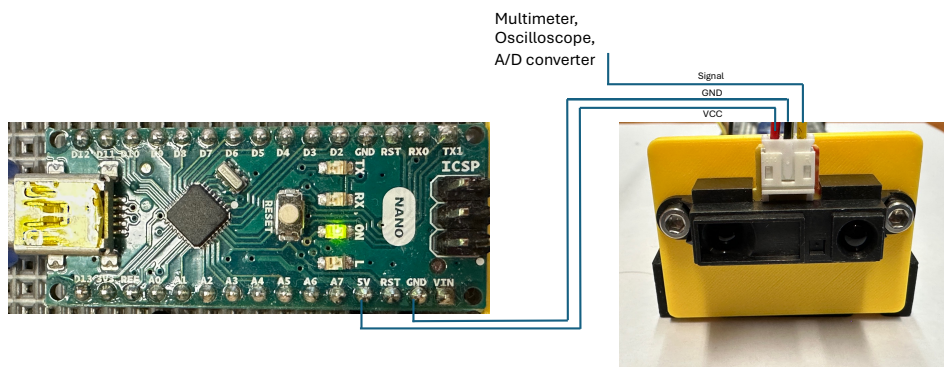
- Optical distance sensing by triangulation and PSD-based sensors
- Sensor characteristics: Measurement range, resolution, accuracy, precision, characteristic curve
- Sensor errors
- Analog voltage interface, A/D-conversion
- Datasheet Sharp GP2Y0A21YK0F IR distance sensor (https://global.sharp/products/device/lineup/data/pdf/datasheet/gp2y0a21yk_e.pdf)
- Measurement tools: Multimeter, Oscilloscope, probe calibration
- Programming an Arduino (Nano) with the Arduino IDE
- Programming in Python (Anaconda/Spyder)
- Python and python libraries (z.B. <https://py-tutorial-de.readthedocs.io/de/python-3.3/>, <https://www.python-kurs.eu/numpy.php>, ...)

Material

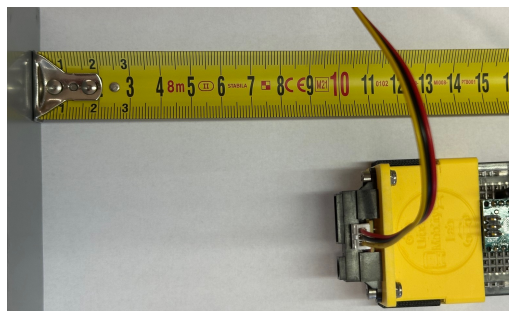
- Arduino Nano
- IR distance sensor Sharp GP2Y0A21YK0F
- Breadboard
- USB-Mini cable
- 3 Male-Male connectors
- Ruler (distance reference)
- Free space with a flat object (wall, cardboard box, ...)
- Laptop, Arduino-IDE, Python-IDE (Anaconda, Spyder)
- In the Lab: Oscilloscope, Voltmeter (Multimeter)

Total time: approx. 1.5 h**Setup**

(a)



(b)



The Arduino is used as the source for 5V/GND, the sensor output is measured with a multimeter (exercise 1), an oscilloscope (exercise 2), and with an A/D-converter of the microcontroller (exercise 3)

Exercise 1 – Sensor Setup, Characteristic Curve, Multimeter Measurements

- Connect the sensor lines as indicated (Supply Voltage, GND, Analog Sensor Output)
- Put a square object in front of the sensor and vary the distance between the sensor front and the object for a range from 0 cm to 50 cm.
- Measure the voltage output of the sensor with a multimeter for at least 20 different distances. Make sure, you chose these values properly to cover all relevant characteristics of the sensor.
- Compare your result with the sensor specification

Deliverables

- Photos of your setup with annotations for the connections and the full view on sensor, object, and the measurement range
- Characteristic curve voltage(distance)
- Assessment – How well does your characteristic curve fit to the curve specified in the sensor documentation? What is the maximum deviation observed?

Exercise 2 – Characteristic Curve, Oscilloscope Measurements

- Calibrate the oscilloscope probes
- Add the oscilloscope to your setup to measure the behavior of the sensor output over time
- What do you observe? What type of problems would you expect when measuring sensor output with a A/D-converter running at about 10 kHz sampling rate?
- Use the oscilloscope math functions to measure the voltage output of the sensor for at least 20 different distances in the range 0 cm to 50 cm. Choose the same values for the distance as in exercise 1.
- Compare your result with the sensor specification and with your result of exercise 1

Deliverables

- Photos of your setup with annotations for the connections and the full view on sensor, object, and the measurement range
- Oscilloscope measurement $u(t)$ and your observation
- Characteristic curve voltage(distance) measured with the oscilloscope
- Assessment – How well does your characteristic curve fit to the curve specified in the sensor documentation and the result of exercise 1? What is the maximum deviation observed?

Exercise 3 – Sensor and Arduino Setup, A/D converter, Time Series

- Setup the Arduino and the IR distance sensor on the breadboard
- Connect three pins of the sensor VCC, GND, Signal to the respective pins on the Arduino (pls. check in the software and adjust values for pins if necessary)
- Load the Arduino code into the IDE, set all parameters for the board, and compile and transfer the code to the microcontroller
- Use “Werkzeuge/Serieller Monitor” or “Werkzeuge/Serieller Plotter” to check the output of the sensor, vary transmission parameters (e.g. Baud rate). What happens, if transmission parameters of Arduino and Computer do not match? Please document your results. What is the maximum Baud-rate?
- Download the python code and open it in Anaconda/Spyder
- Set all transmission parameters correctly (port, baud rate)
- Start the program and check the values printed on the screen. You may have to install the libraries „pyserial“, „numpy“ and „matplotlib“

Deliverables

- Annotated photo of your setup (label all electrical connections, include the object, and the measured distance)
- Block diagram and data flow diagram for your setup
- Documentation and description of the consequences of baud-rates not matching between the Arduino and the receiving program
- Time series, histogram, and statistical indicators (mean, median, standard deviation) for a measurement distance of 10 cm with at least 1000 measured values

- Description: What do you observe on the time series of the sensor output? What are the consequences? What countermeasure would you propose?

Exercise 4 – Characteristic Curve, A/D-Converter

- Setup the A/D-converter and include your corrective action into the code to prepare for the measurement in this exercise.
- Use the A/D-converter to measure the voltage output of the sensor for at least 20 different distances in the range 0 cm to 50 cm. Choose the same values for the distance as in exercises 1 and 2.
- Compare the characteristic curve of this sensor from the specification to your results in the exercises 1, 2, and 4.

Deliverables

- Your code including the countermeasure, explanation of the functionality
- Characteristic curve of your sensor based on the A/D-converter
- Overall assessment of the characteristic curve of this sensor from the specification and your results of the exercises 1, 2, and 4.